

KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

*Enabling agents to autonomously decide when to act directly,
self-reflect, or consult external knowledge — only when needed.*

Fast Thinking

Slow Thinking

Knowledgeable Thinking

Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge Injection

① Trajectory Imitation

ReAct-style

Blindly mimics gold trajectories.
Fragile on unexpected inputs.

② Trial-and-Error Refinement

Reflexion-style

External feedback injected at
every step regardless of need.

③ Knowledge-Augmented Planning

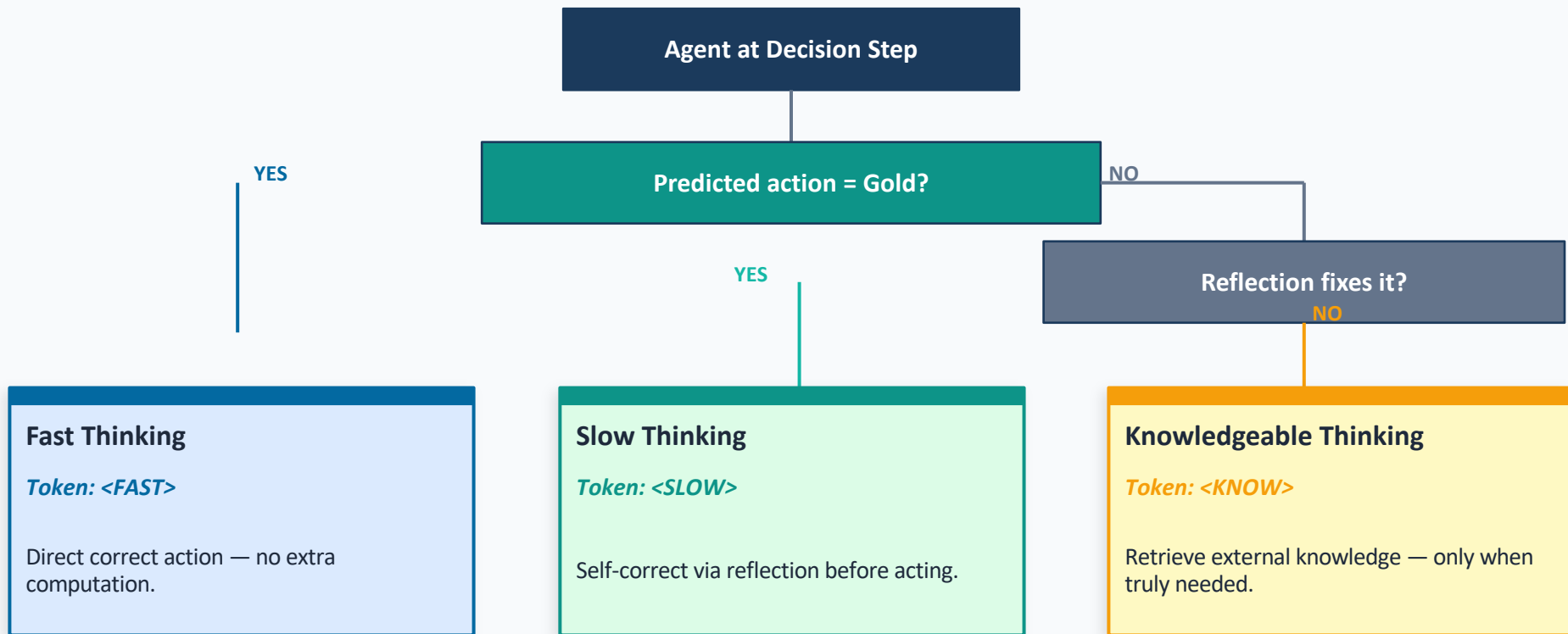
KnowAgent / WKM

Domain knowledge always prepended —
100% of steps, unconditionally.

Result: Wasteful inference, pattern collapse, and suboptimal performance on novel inputs.

Humans use situational self-awareness — assessing whether to act directly, pause and reflect, or seek outside help. KnowSelf brings this metacognitive ability to LLM agents.

Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking



KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

01

Self-Awareness Data Construction

- Agent self-explores environments
- Heuristic labels each step:
Predicted = Gold → <FAST>
Reflection fixes it → <SLOW>
Neither works → <KNOW>
- Special tokens inserted into trajectory

02

Two-Stage Training

- Stage 1 — Supervised Fine-Tuning (SFT):
Learns initial self-awareness patterns
- Stage 2 — RPO (Reward-based Policy Optimization):
Refines special-token generation
Reward based on correct situation ID
No expensive human annotation needed

03

Self-Aware Inference

- Agent generates situation token each step
- <FAST> → execute direct action
- <SLOW> → trigger self-reflection loop
- <KNOW> → query external knowledge base
- Result: only ~15–26% of steps need knowledge retrieval

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Task A: "Put a hot mug in cabinet"

OpenAI-O1 (Baseline) ✗

Opens microwave — then incorrectly tries to take an egg from it. Misidentifies the object; task fails.

KnowSelf (Llama-8B) ✓

Generates <SLOW> Reflection token — realizes it should heat the mug using the already-open microwave. Task succeeds.

Task B: "Put a saltshaker in drawer"

DeepSeek-R1 (Baseline) ✗

Reflects that the drawer is open and tries to close it first. Action fails — drawer cannot be closed with items in the way. Gets stuck.

KnowSelf (Llama-8B) ✓

Generates <KNOW> Knowledge token — retrieves: place the object without removing unrelated items. Task succeeds.

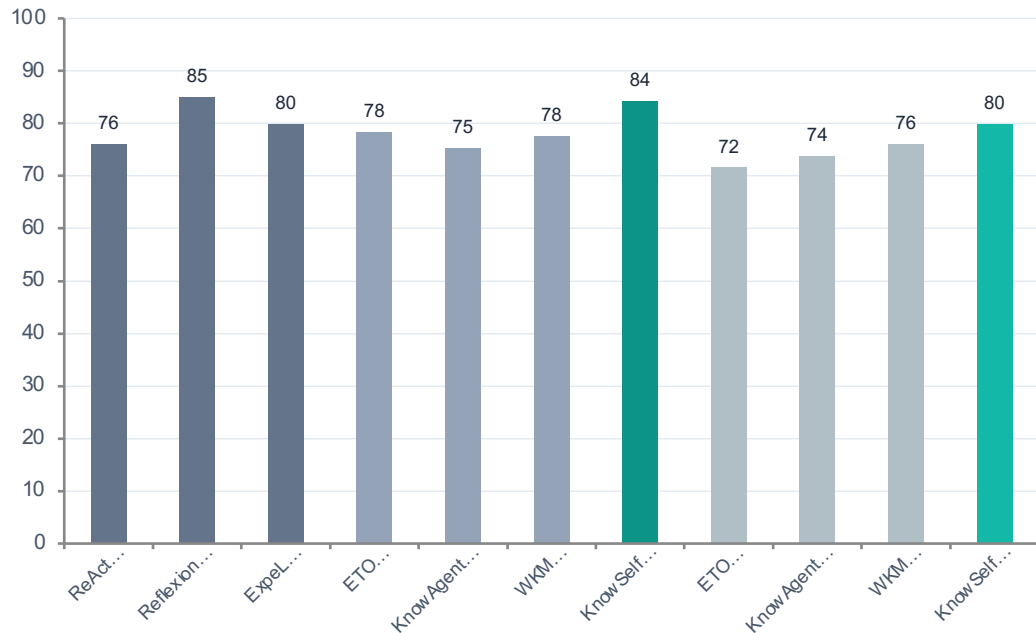
KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All (%)
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

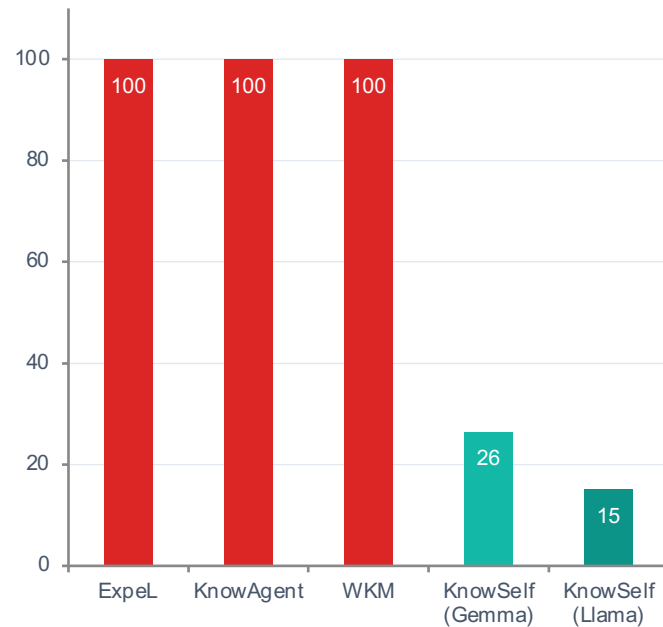
Highlighted rows = KnowSelf. Red Know% = 100% (indiscriminate). Teal Know% = selective (15–26%). All = overall success rate.

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage

Overall Accuracy (%) by Method



Knowledge Usage (%) by Method



KnowSelf uses only 15–26% knowledge queries yet achieves the highest accuracy among open-weight models — beating methods that always retrieve knowledge (100%).

Key Takeaways

Self-Awareness Is the Missing Piece in Agent Planning

1

Indiscriminate knowledge injection (flood irrigation) is wasteful — using 100% knowledge can hurt more than it helps.

2

Agents can learn situational self-awareness via lightweight heuristic labeling and special tokens — no expensive annotation needed.

3

KnowSelf (Llama-8B) achieves 84.33% on ALFWorld, outperforming GPT-4o ReAct (76.12%) and GPT-4o ExpEL (79.85%).

4

Only 15–26% of steps require external knowledge, vs. 100% for KnowAgent / WKM / ExpEL — a 4–7× reduction in retrieval cost.

5

Generalizes across scales: Gemma-2B reaches 79.85%, matching GPT-4o ExpEL — no frontier model required.